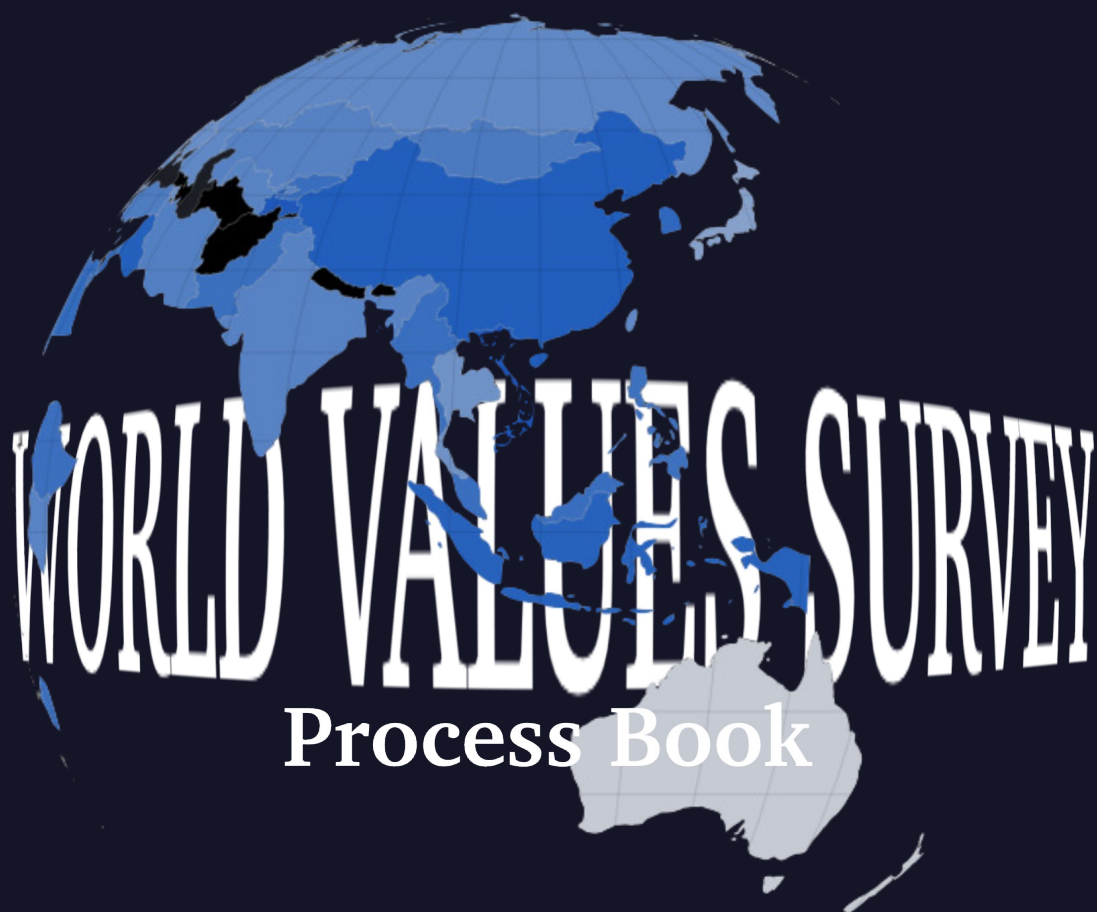




Introduction

Have you ever wondered how well you might fit in if you moved abroad?

We did, and that's where this project began.

Our goal was to quantify how smoothly someone could integrate into a new country. To do this, we analyzed the World Values Survey, focusing on questions about trust, tolerance, and friendship to calculate a “cultural compatibility” score for each nation.

But we didn't stop at a single overall score. We know that different groups (women, various age groups, religious communities, and income levels) have unique perspectives. That's why we offer multiple demographic filters, an interactive world map highlighting your best-fit countries, and detailed breakdowns of each country's survey responses by subgroup.

Data Selection and Preprocessing

The project began with selecting the right dataset and then processing it to meet our requirements. We chose the World Values Dataset, wave 7 (2017-2022). The dataset consists of answers to questions around various subjects, ranging from social values and attitudes to political interest and security.

We decided to focus on specific areas, reducing the number of questions to handle. A dozen questions were selected for each of the selected categories.

We then had to ensure the questions we ask the user would follow the same structure as the questionnaire and thus created an “overall question” and “specific question” (e.g. Overall question: “How do you feel about...” Specific question: “Homosexuality”/ “Family” etc.), and linked these to the corresponding question ID in the dataframe (e.g. Q1,Q2, etc.).

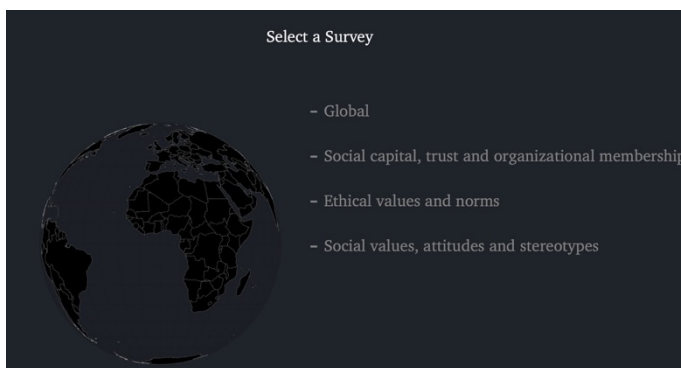
Another challenge was that the amount of answers varies from one question to the next (e.g. ‘Strongly agree’, ‘Agree’, ‘Disagree’, ‘Strongly Disagree’ or ‘1’,‘2’,‘3’,...,‘10’).

This necessitated for us to store the possible answers in a dictionary mapping the answer in the dataframe to the corresponding label (e.g. ‘1’ = ‘Strongly agree’).

Questions are also linked to the different quiz we allow (Global; Social values, Attitudes and Stereotypes, etc.)

Extracting the questions from the dataset was rather difficult, as we initially tried parsing the PDF to extract the questions/answers, but the structure did not allow to do so and we turned to manual extraction with help from ChatGPT. This resulted in the file `handwritten_answers.csv`.

Main page



The site's main page features two panels: a globe visual on the left and the survey and visualizations on the right. As users answer questions, we dynamically update the globe's heatmap to reflect their evolving compatibility scores.

To achieve this, each new response triggers a recalculation of similarity scores, which feeds into the heatmap generation.

Survey Answers Comparison

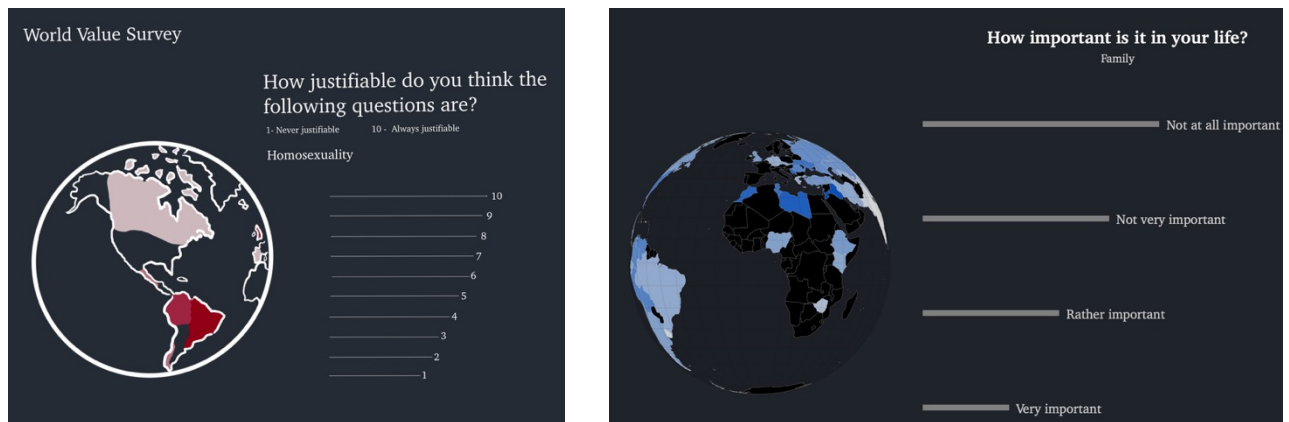
While comparing survey responses is a complex statistical challenge, our primary goal was to create a website that clearly illustrates the diversity of societal values across the globe. This required a similarity metric that is both easy to interpret and visually display across countries. We chose the Mean Absolute Error (MAE) as our core metric, given its suitability for ordinal data and its intuitive meaning. For future work, alternative approaches such as cosine distance could be explored, particularly if the number of selected questions increases and more nuanced comparisons are needed.

Once the MAE is computed for each respondent, we group these per country and take the mean, which is ultimately our score.

Once a question is answered, we add the resulting score to the previous ones (from previous answers) and take the mean.

Note that computing the distance required dynamic preprocessing, which added some difficulty to this step.

Dynamic Question and Topics Panels



We first extracted the selected survey, allowing to take the corresponding questions, and kept only a subset of the questions.

Out of the selected questions, we dropped all users that didn't answer at least one of our remaining questions, which allowed us to keep as many users as possible.

We could have dropped these users during the data preprocessing step, but felt this would have significantly reduced the quality of our analysis.

The questions are different each time, making the quiz more fun for the user.

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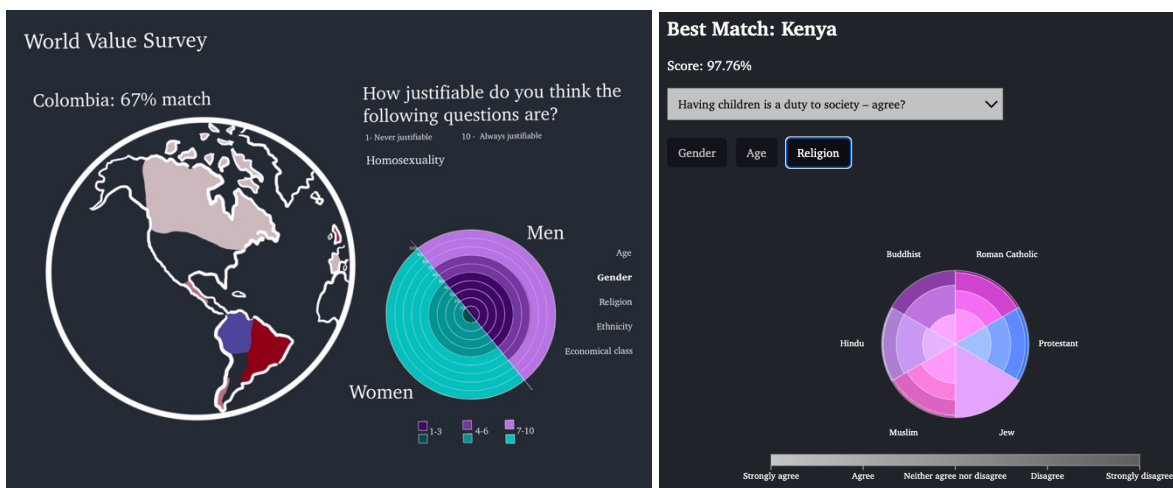
To achieve this, each new response triggers a recalculation of similarity scores, which feeds into the heatmap generation.

Quiz Results per Country

After the user is done taking the quiz, we wanted to first render the best match, and also give the user the liberty to go deeper into the analysis.

We felt that an overall score was too general, and it would be very relevant to highlight what different groups within a country answered. Indeed, an 18 year-old person might care more what people their age answered.

Grouping the answer for each group proved time-consuming, and we opted only to select the Age groups, Sexes, and religions



Building the histograms

The visualization is a revisited version of the stacked histogram, which we made using [D3.js](https://d3js.org/).

since the questions are organized in two parts ("Overall question" and "Specific question"), we needed to combine these two into one question on the dropdown menu.

We dynamically processed the dataset by selecting all users from the selected country (default set to best match), removing all users which did not answer all explorable questions and removing respondents who did not answer the demographics questions (age, religion or gender).

This was again in order to visualise as many respondents as possible. We observed that some subgroups have very little respondents, which can introduce some bias.

Legend

We used a grey color gradient legend, which could generalize to any palette.

We wanted to plot the different color gradients for each group as in our original sketches, but there would have been too many palettes to display since religion can be split into 8 different groups, thus leading to 8 different color gradients

Why stop at the best match?

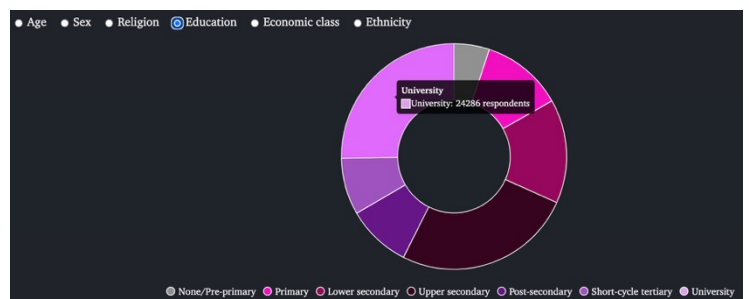
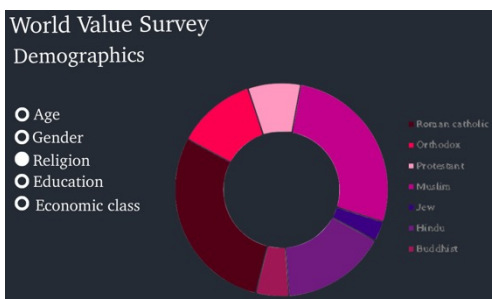
We decided to allow the user to analyze any country on the globe, displaying the match score as well as the answers for this specific country.

This allows users to not only explore the 'best country' according to our algorithm, but also other highly matching countries.

Explore page

In order to allow the user to know more about the dataset and the people that answered the questions which ultimately lead to the score, we decided to add an 'Explore page'.

Donut Chart



We first decided to display a standard **Donut Chart** to show the distributions in Age, Sex, Religion, Education, Economic Class, and Ethnicity. We use Chart.js for this purpose.

A first challenge was to define the categories to keep, as these may vary from country to country.

For the Age, we defined the age groups as follows: 18-24, 25-34, 35-44, 45-54, 55-64, 65+. We then iterated over the column corresponding to the age (Q262) and counted the individuals in each category, with a dedicated function (getAgeCounts).

For the Sex, Religion, Education, and Economic Class, the approach is similar.

However, for the Ethnicity, we had to decide which main ethnic groups to keep. We tried to choose a selection as diverse as possible, but since many countries had their own specific ethnic groups, which did not fit in our pre-defined groups, we ended up having a lot of individuals classified as 'Others' (e. g. 'light brown', 'dark brown' etc.). The ethnicity map is in the file ethnicity_map.csv, which was done manually and with some help of ChatGPT.

Similarities between Countries



We then wondered how countries' populations might be similar overall. That is, if we take an average answer for every country, how could we find an intuitive and simple representation of how countries are similar to each other in terms of the values of their inhabitants.

We agreed on t-SNE to visualize countries as 2D bubbles. We only kept the relevant questions, meaning those available for the surveys. This made the dataset much smaller and made more sense since we decided to focus on these few questions.

We then computed the mean of each of the selected questions for each country, and used t-SNE to visualize the results in 2D.

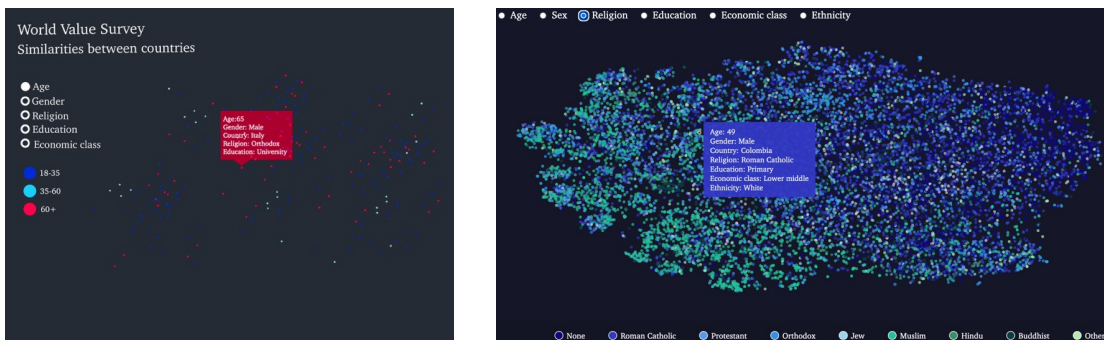
The clustering is performed in the file clusters.ipynb, and the chart uses D3.js.

Similarities between Individuals

We decided another t-SNE plot to visualize individuals.

This was very challenging as we wanted to be able to highlight bubbles with different colors to show groups of different people, to see if potential clusters were explained by the nationality or other factors.

The t-SNE plot was easy to generate since we had the experience of the previous graph, but the real difficulty came when trying to highlight bubbles in different colors. We overcame the challenge by first creating a comprehensive color mapping system for each demographic category, then joining the t-SNE embeddings with demographic data (sampling 30% of individuals to prevent overcrowding), and finally implementing a dynamic D3.js visualization that updates both bubble colors and the legend whenever users switch between demographic attributes using radio buttons, while maintaining interactive tooltips that show the complete demographic profile of each individual. The result is displayed in the figure below.



Peer Assessment:

- ✧ Website scaffold: Ensuring different browsers compatibility/ Globe fonctionnalites/ interactive bars: **Jérémy**
- ✧ Exploratory Data Analysis/ General data preprocessing: **Camille**
- ✧ Country matching functionality and per country histogram analysis: **Camille**
- ✧ Explore page chart preprocessing and functionalities (Donut charts, countries bubble map, individuals bubble map and the t-SNE): **Renuka**
- ✧ Website design (font, colors): **Jérémy** and **Renuka**